

PLAIN-LANGUAGE GUIDE

RegenAI

A digital assistant that helps farmers find conservation money, earn from carbon, and know what to do next on their fields.



What it is · Who it's for · How it works · Where it stands today
September 2026

1. The idea in one minute

Farmers in the U.S. Midwest can get paid for farming in ways that protect the soil and water. This is called **regenerative** or **conservation** farming, and it includes practices like planting cover crops over winter or not plowing the ground.

The money is real, but it's hard to get to. It comes from government programs with confusing rules, deadlines that differ by state, and scoring worksheets. Companies that buy **carbon credits** are another source, and also hard to navigate. Most farmers don't have time to work all of that out.

RegenAI works it out for them. A farmer describes their farm once. The app then tells them which programs they probably qualify for, how much they could earn, and what practical steps would earn them more.

Who it's for

Family and commercial farms of **500 to 5,000 acres** growing row crops like corn and soybeans. They're in nine states: Illinois, Indiana, Iowa, Kansas, Minnesota, Missouri, Nebraska, Ohio and Wisconsin. The app is built to work on a phone out in the field, with big buttons and plain wording.

The business

Farms pay a monthly subscription of about **\$99–\$299**. The first goal is to sign up **5 paying pilot farms**.

2. What the app does for a farmer

"What should I do this week?"

The app uses AI to suggest 3–6 practical steps, like "plant a cereal rye cover crop on the North Field." Each one comes with a plain-English reason.

"How's my field doing?"

It pulls in the weather forecast and official government soil data for each field automatically.

"Am I eligible for programs?"

It checks the farm against two big USDA programs, EQIP and CSP. It estimates payments and shows application deadlines.

"Can I earn carbon credits?"

It estimates how many carbon credits the farm's practices could generate.

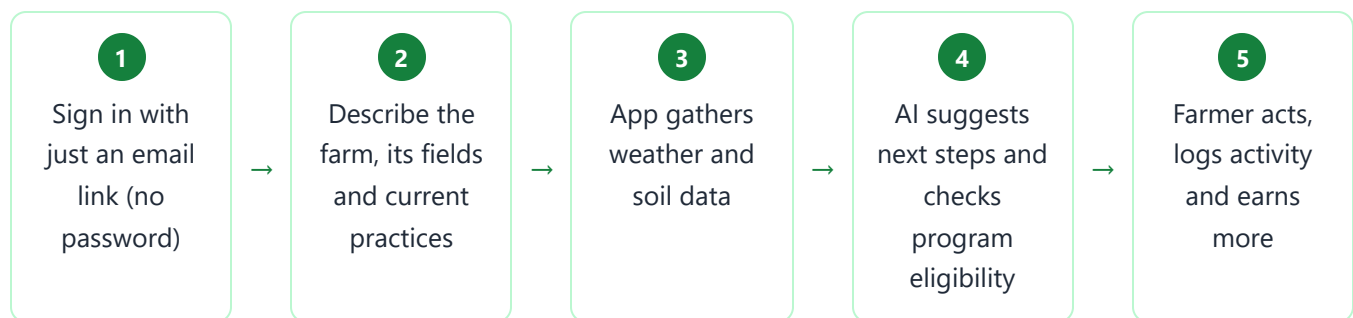
"What happened recently?"

It's a digital field notebook for planting, spraying, harvest and yields. It also calculates the farm's official average yield, which crop insurance uses.

Paperwork

Farmers can upload soil reports, photos and compliance documents that programs ask for as proof.

3. A farmer's journey



Every time the farmer marks a suggestion as done, their program eligibility and carbon estimates can go up. That gives them a reason to keep coming back.

4. How it works behind the scenes

You can think of the app as three layers:

Layer	What it does	Technology
The screens what the farmer sees	Web pages that work well on phones: the dashboard, sign-up steps, activity log and program pages.	Next.js, React, Tailwind
The brain the server	Runs all the calculations (program scoring, payments, carbon estimates), talks to the AI, and fetches weather and soil data.	Python, FastAPI
The memory the database	Stores farms, fields, suggestions and activity. It's locked down so each farmer can only see their own farm.	Supabase (PostgreSQL)

Outside services it talks to

- **Claude (Anthropic's AI)** writes the farming suggestions.
- **Open-Meteo** provides weather forecasts.
- **USDA soil survey (SSURGO)** provides official soil information for each field's location.

Keeping the AI honest. AI can make things up. RegenAI checks every suggestion before a farmer sees it. The conservation practice has to be a real, official USDA practice code, and the field it mentions has to actually belong to that farm. Anything that fails the check is thrown out.

5. The money programs, simply explained

EQIP: help paying for new practices

EQIP is a USDA program that pays part of the cost when a farmer adopts a conservation practice. RegenAI matches what the farmer has done against the official list of EQIP practices. It also checks that they have some supporting paperwork.

CSP: rewards for good stewardship

CSP pays farmers who already take good care of their land, and pays more for doing even better. RegenAI scores the farm on **8 areas**, including soil health, erosion, water quality, air quality and energy, for up to 100 points in total. It then gives one of four answers:

Result	What it means
Act now	The farm scores high enough for a fast-track approval. It should apply soon.
Eligible	The farm qualifies, and a few more practices could move it to fast-track.
Pending review	The farm is close. It meets the bar in one area and needs two.
Not eligible yet	The farm should start with some basic practices. The app shows which ones.

It also estimates the payment: at least **\$4,000** and at most **\$50,000 a year**, up to **\$200,000** over a 5-year contract.

Carbon credits

Some companies pay farmers for storing carbon in their soil. RegenAI estimates credits per acre for three practices: **cover crops**, **no-till** and **crop rotation**. Healthier soil, meaning more organic matter, gets a higher estimate.

6. Where the project stands today

RegenAI is an **early working version** (an "MVP"). It's been through two development sprints. The calculations and the AI pipeline are built and covered by about 400 automated tests.

A September 2026 review found parts that **weren't connected yet**. These have now been fixed:

- The screens and the server now ask each other for information in the same format.
- Sign-up now really saves the farm and its fields, then starts the analysis.
- The dashboard's weather and soil panels show real data.
- A setup error that stopped the dashboard from loading is fixed.
- Automatic checks now run on every change, and all the tests pass.
- Mislabelled conservation practice codes in the reference data are corrected.

Still to do:

- Update the program deadlines, which are last year's.
- Save the enhancement choices a farmer picks.
- Extend soil and weather lookups to more counties.
- Double-check a few practice codes and all program rules and carbon rates against the latest official USDA and carbon-market figures before real farmers rely on them.

7. Words you might hear

Regenerative agriculture

Farming in ways that rebuild soil health and protect water, rather than wearing the land down.

Cover crop

A crop, such as cereal rye, planted between cash-crop seasons to protect and feed the soil.

No-till

Planting without plowing, which keeps soil and carbon in the ground.

USDA / NRCS

The U.S. Department of Agriculture, and its office that runs conservation programs.

EQIP

Environmental Quality Incentives Program. It helps pay for adopting conservation practices.

CSP

Conservation Stewardship Program. It pays farmers for ongoing good land care.

CART score

The USDA's scoring system for deciding which farms get CSP contracts.

Carbon credit

A certificate showing that one tonne of CO₂ was kept out of the air. Companies buy these.

APH (Actual Production History)

A farm's official average yield over past years, used for crop insurance.

MVP

Minimum Viable Product: the first version with just enough features to test with real customers.